Assignment V

1. **Finding Unitary Matrix by Hand**
   Construct the matrix for the transformation from the basis of σ_y to σ_x. In other words, finding the unitary matrix to re-represent a vector in the σ_y eigenvector basis representation in the σ_x eigenvector basis representation.

2. **Completeness of Basis**
   Show that the eigenvectors of σ_x form a complete basis (Hint: see the following equation).

$$\sum_{i=0}^{n-1} |i\rangle\langle i| = |-\rangle\langle -| + |+\rangle\langle +|$$
$$= \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}1 & -1\end{pmatrix} + \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\end{pmatrix}$$
$$= \frac{1}{2}\begin{pmatrix}1 & -1\\-1 & 1\end{pmatrix} + \frac{1}{2}\begin{pmatrix}1 & 1\\1 & 1\end{pmatrix} = \begin{pmatrix}1 & 0\\0 & 1\end{pmatrix} = I$$

3. **Construction of Matrix**
   Construct the σ_x by using its eigenvalues and eigenvectors.

4. **Colab Verification**
   Use Colab to verify the results from Problems 1 to 3.

5. **Computing Tensor Product**
   Let $|f\rangle = \begin{pmatrix}0\\1\end{pmatrix}$ and $|g\rangle = \begin{pmatrix}\frac{1}{\sqrt{2}}\\\frac{i}{\sqrt{2}}\end{pmatrix}$, find $|f\rangle_1 \otimes |g\rangle_2$ and $|g\rangle_1 \otimes |f\rangle_2$. Are they the same? Here you see the order is very important.

6. **Computing Tensor Products of Three Vectors**
   Find $\begin{pmatrix}0\\1\end{pmatrix} \otimes \begin{pmatrix}\frac{1}{\sqrt{2}}\\\frac{i}{\sqrt{2}}\end{pmatrix} \otimes \begin{pmatrix}0\\1\end{pmatrix}$. What is the dimension of this space?

7. **Distribution Law**
   Check the following equation:

$$(|f\rangle + |e\rangle) \otimes |g\rangle = |f\rangle \otimes |g\rangle + |e\rangle \otimes |g\rangle$$
$$|g\rangle \otimes (|f\rangle + |e\rangle) = |g\rangle \otimes |f\rangle + |g\rangle \otimes |e\rangle$$

by substituting $|f\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, $|e\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $|g\rangle = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} \end{pmatrix}$ to both sides.

8. Tensor Products using Colab

Use Colab to verify the results from Problems 5 to 7. For Problem 5, you may use this code

```
import numpy as np
g = np.array([[0],[1]])
h = np.array([[1/np.sqrt(2)],[1j/np.sqrt(2)]])
print(np.kron(g,h))
print(np.kron(h,g))
```

9. Inner Product of Tensor Products

Check the following equation:

$$\langle h_1|h_2\rangle = \left\langle \sum_{j=1}^{n_1} |f_i\rangle \otimes |g_j\rangle \middle| \sum_{l=1}^{n_2} |e_l\rangle \otimes |k_l\rangle \right\rangle = \sum_{j=1}^{n_1} \sum_{l=1}^{n_2} \langle f_j|e_l\rangle \langle g_j|k_l\rangle$$

but setting $n_1 = n_2 = 1$ (i.e. only one term) and $|f\rangle = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, $|e\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, $|g\rangle = \begin{pmatrix} i \\ i \end{pmatrix}$, and $|k\rangle = \begin{pmatrix} i \\ 1 \end{pmatrix}$.

10. Hand Calculation of Tensor Product

Let $|f\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ and $|g\rangle = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} \end{pmatrix}$, find $\sigma_x|f\rangle_1$ and $\sigma_x|g\rangle_2$. Then find the tensor product of the resulting vector (i.e. $\sigma_x|f\rangle_1 \otimes \sigma_x|g\rangle_2$). Now find $|f\rangle_1 \otimes |g\rangle_2$ and construct $\sigma_x \otimes \sigma_x$. Show that $\sigma_x \otimes \sigma_x(|f\rangle_1 \otimes |g\rangle_2)$ gives the same result.

11. Tensor Product Calculation using Colab

Use Colab and the function np.kron to verify Problem 10.